



الجامعة الإسلامية العالمية ماليزيا
INTERNATIONAL ISLAMIC UNIVERSITY MALAYSIA
يُونُسِيَّتِيْ اِسْلَامُ اِنْتَارَا بَغْسَا مِلْسِيَا

Garden of Knowledge and Virtue

INFO 2303 DATABASE PROGRAMMING

SEM 1, 2023/2024

SECTION 1

GROUP PROJECT FINAL REPORT

PROJECT TITLE:

Toilet Cleaning Service Booking System

GROUP MEMBERS:

Name	Matric No
NUR WARDAH FATIMAH BINTI SAFUAN	2210902
NUR ALIFAH ILYANA BINTI MOHD ZAMAN	2217466
ADLYN SOFEA BINTI MOHD ARIFF ADIKA	2214042
BIDAYATUN NAJIHAH BINTI BUKHAARI	2216798

INSTRUCTOR: Dr. Zainatul Shima Abdullah

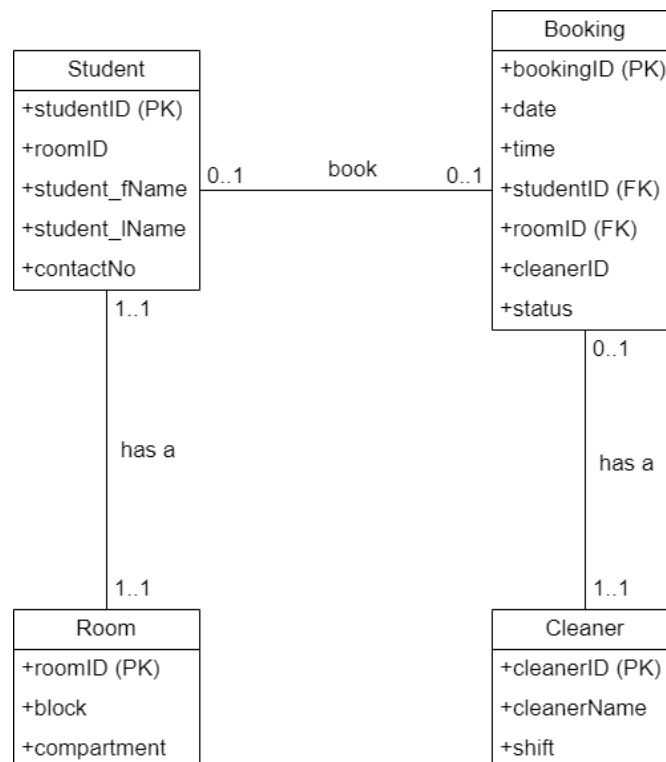
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1. Introduction

Environmental protection is a global issue that is gaining increasing attention around the world. In Islam, caring for the environment is the responsibility of all mankind. In International Islamic University Malaysia (IIUM), we believe that maintaining cleanliness is a shared responsibility within the IIUM community. Therefore, our group came up with a new Toilet Cleaning Service Booking System called "AquaShine". Compared to the traditional WhatsApp method, our system can provide more efficiency and convenience for all users. No more juggling messages and schedules on WhatsApp. Our intended users are admins and staff (Mahallah Officers). Students, living in Mahallah Aminah, can easily schedule toilet cleaning services through the system, ensuring a hygienic environment. Cleaners have a centralised platform to manage their schedules and assignments efficiently. Mahallah Officers can utilise the system to monitor cleanliness standards, track cleaning activities, and address specific requirements within their respective Mahallahs. This comprehensive system enhances collaboration, transparency, and hygiene management across the entire IIUM community, fostering a clean and conducive living for everyone.

2. Entity Relationship Diagram (ERD)



3. Data Dictionary

CLEANER

ATTRIBUTE	DATA TYPE	SIZE	NULL
cleanerID	NUMBER	3	NO (PK)
cleaner_name	VARCHAR2	50	NO
shift	VARCHAR2	10	NO

STUDENT

ATTRIBUTE	DATA TYPE	SIZE	NULL
studentID	NUMBER	7	NO (PK)
student_fname	VARCHAR2	50	NO
student_lname	VARCHAR2	50	NO
contact_no	VARCHAR2	15	NO
roomID	VARCHAR2	5	NO (FK)

ROOM

ATTRIBUTE	DATA TYPE	SIZE	NULL
roomID	VARCHAR2	5	NO (PK)
block_name	CHAR	-	NO
room_level	NUMBER	2	NO
compartment	CHAR	-	NO

room_status	VARCHAR2	10	YES
-------------	----------	----	-----

BOOKING

ATTRIBUTE	DATA TYPE	SIZE	NULL
bookingID	NUMBER	3	NO (PK)
booking_date	DATE	-	NO
booking_time	TIME	-	NO
booking_status	VARCHAR2	10	NO
studentID	NUMBER	7	NO (FK)
roomID	VARCHAR2	5	NO (FK)
cleanerID	NUMBER	3	NO (FK)

4. Data Definition Language (DDL) - Create Table

DROP TABLE STUDENT cascade constraints;

DROP TABLE CLEANER cascade constraints;

DROP TABLE ROOM cascade constraints;

DROP TABLE BOOKING cascade constraints;

```
CREATE TABLE STUDENT (  
    studentID NUMBER(7),  
    student_fname VARCHAR(50) NOT NULL,  
    student_lname VARCHAR(50) NOT NULL,  
    contact_no VARCHAR(15) NOT NULL,  
    roomID VARCHAR(5) NOT NULL,  
    CONSTRAINT STUDENT_studentID_PK PRIMARY KEY (studentID)  
);
```

```
CREATE TABLE CLEANER (  
    cleanerID NUMBER(3),  
    cleaner_name VARCHAR(50) NOT NULL,  
    shift VARCHAR(10) NOT NULL,  
    CONSTRAINT CLEANER_cleanerID_PK PRIMARY KEY (cleanerID)  
);
```

```
CREATE TABLE ROOM (  
    roomID VARCHAR2(5),  
    block_name CHAR NOT NULL,  
    room_level VARCHAR(2) NOT NULL,  
    compartment CHAR NOT NULL,  
    room_status VARCHAR(10),  
    CONSTRAINT ROOM_roomID_PK PRIMARY KEY (roomID)
```

);

CREATE TABLE BOOKING (

 bookingID NUMBER(3),

 booking_date DATE,

 booking_time DATE,

 booking_status VARCHAR2(10) NOT NULL,

 studentID NUMBER(7) NOT NULL,

 roomID VARCHAR2(5) NOT NULL,

 cleanerID NUMBER(3),

 CONSTRAINT BOOKING_bookingID_PK PRIMARY KEY (bookingID),

 CONSTRAINT BOOKING_studentID_FK FOREIGN KEY (studentID)

REFERENCES STUDENT,

 CONSTRAINT BOOKING_roomID_FK FOREIGN KEY (roomID)

REFERENCES ROOM,

 CONSTRAINT BOOKING_cleanerID_FK FOREIGN KEY (cleanerID)

REFERENCES CLEANER

);

5. Data Definition Language (DDL) - Data Population

--table student

```
INSERT INTO student VALUES (2214301, 'Sabrina', 'Melati', '60173694852',  
'A2.2D');
```

```
INSERT INTO student VALUES (2223692, 'Dahlia', 'Alya', '60143357445', 'C1.3A');
```

```
INSERT INTO student VALUES (2123389, 'Batrisyia', 'Awatif', '60195858632',  
'B2.1A');
```

```
INSERT INTO student VALUES (1920443, 'Ain', 'Adlin', '60122235456', 'H5.3D');
```

```
INSERT INTO student VALUES (1926232, 'Suria', 'Ilyana', '60133546191', 'E3.2B');
```

```
INSERT INTO student VALUES (2015486, 'Puteri', 'Najihah', '60102547885',  
'G1.2C');
```

```
INSERT INTO student VALUES (2025445, 'Aliyah', 'Fatimah', '60135945542',  
'E4.1D');
```

```
INSERT INTO student VALUES (2312241, 'Adlina', 'Sofia', '60144556744', 'E2.2A');
```

```
INSERT INTO student VALUES (1912346, 'Balqis', 'Raihana', '60103235965',  
'H3.4C');
```

```
INSERT INTO student VALUES (2228975, 'Aisya', 'Amirah', '60136964223',  
'A1.1A');
```

```
INSERT INTO student VALUES (2016583, 'Alisya', 'Khadija', '60177455854',  
'G4.4B');
```

```
INSERT INTO student VALUES (2217611, 'Maryam', 'Aminah', '60113432556',  
'B2.3A');
```

```
INSERT INTO student VALUES (2210923, 'Hafsa', 'Halimah', '60111137695',  
'D2.4B');
```

```
INSERT INTO student VALUES (2219725, 'Nur', 'Irina', '60133566442', 'H5.3A');
```

```
INSERT INTO student VALUES (1924733, 'Putri', 'Sarah', '0113255669', 'A4.3C');
```


	STUDENTID	STUDENT_FNAME	STUDENT_LNAME	CONTACT_NO	ROOMID
1	2214301	Sabrina	Melati	60173694852	A2.2D
2	2223692	Dahlia	Alya	60143357445	C1.3A
3	2123389	Batrisyia	Awatif	60195858632	B2.1A
4	1920443	Ain	Adlin	60122235456	H5.3D
5	1926232	Suria	Ilyana	60133546191	E3.2B
6	2015486	Puteri	Najihah	60102547885	G1.2C
7	2025445	Aliyah	Fatimah	60135945542	E4.1D
8	2312241	Adlina	Sofia	60144556744	E2.2A
9	1912346	Balqis	Raihana	60103235965	H3.4C
10	2228975	Aisya	Amirah	60136964223	A1.1A
11	2016583	Alisya	Khadija	60177455854	G4.4B
12	2217611	Maryam	Aminah	60113432556	B2.3A
13	2210923	Hafsa	Halimah	60111137695	D2.4B
14	2219725	Nur	Irina	60133566442	H5.3D
15	1924733	Putri	Sarah	0113255669	A4.3C

--table cleaner

INSERT INTO cleaner VALUES (101, 'Salmah', 'Morning');

INSERT INTO cleaner VALUES (102, 'Rokiah', 'Noon');

INSERT INTO cleaner VALUES (103, 'Fatin', 'Evening');

INSERT INTO cleaner VALUES (104, 'Saripah', 'Morning');

INSERT INTO cleaner VALUES (105, 'Khalida', 'Noon');

INSERT INTO cleaner VALUES (106, 'Kamariah', 'Evening');

INSERT INTO cleaner VALUES (107, 'Azi', 'Morning');

INSERT INTO cleaner VALUES (108, 'Sarina', 'Noon');

INSERT INTO cleaner VALUES (109, 'Fida', 'Evening');

INSERT INTO cleaner VALUES (110, 'Rahmah', 'Morning');

INSERT INTO cleaner VALUES (111, 'Hani', 'Noon');

	CLEANERID	CLEANER_NAME	SHIFT
1	101	Salmah	Morning
2	102	Rokiah	Noon
3	103	Fatin	Evening
4	104	Saripah	Morning
5	105	Khalida	Noon
6	106	Kamariah	Evening
7	107	Azi	Morning
8	108	Sarina	Noon
9	109	Fida	Evening
10	110	Rahmah	Morning
11	111	Hani	Noon

--table room

```

INSERT INTO room VALUES ('B2.1A', 'B', '1', 'A', 'Occupied');
INSERT INTO room VALUES ('G1.2C', 'G', '1', 'C', 'Occupied');
INSERT INTO room VALUES ('H3.3A', 'H', '3', 'A', 'Vacant');
INSERT INTO room VALUES ('E2.2A', 'E', '2', 'A', 'Occupied');
INSERT INTO room VALUES ('D2.4B', 'D', '2', 'B', 'Vacant');
INSERT INTO room VALUES ('F1.2A', 'F', '1', 'A', 'NULL');
INSERT INTO room VALUES ('F2.3C', 'F', '2', 'C', 'NULL');
INSERT INTO room VALUES ('H5.3D', 'H', '5', 'D', 'Occupied');
INSERT INTO room VALUES ('G5.2B', 'G', '5', 'B', 'Occupied');
INSERT INTO room VALUES ('A4.4B', 'A', '4', 'B', 'Occupied');
INSERT INTO room VALUES ('A3.1A', 'A', '3', 'A', 'Vacant');
INSERT INTO room VALUES ('A2.2D', 'A', '2', 'D', 'Occupied');
INSERT INTO room VALUES ('C1.3A', 'C', '1', 'A', 'Occupied');
INSERT INTO room VALUES ('E3.2B', 'E', '3', 'B', 'Occupied');
INSERT INTO room VALUES ('E4.1D', 'E', '4', 'D', 'Occupied');
INSERT INTO room VALUES ('H3.4C', 'H', '3', 'C', 'Occupied');
INSERT INTO room VALUES ('A1.1A', 'A', '1', 'A', 'Occupied');
INSERT INTO room VALUES ('G4.4B', 'G', '4', 'B', 'Occupied');
INSERT INTO room VALUES ('B2.3A', 'B', '2', 'A', 'Occupied');

```

INSERT INTO room VALUES ('A4.3C', 'A', '4', 'C', 'Occupied');

ROOMID	BLOCK_NAME	ROOM_LEVEL	COMPARTMENT	ROOM_STATUS
1 B2.1A	B	1	A	Occupied
2 G1.2C	G	1	C	Occupied
3 H3.3A	H	3	A	Vacant
4 E2.2A	E	2	A	Occupied
5 D2.4B	D	2	B	Vacant
6 F1.2A	F	1	A	NULL
7 F2.3C	F	2	C	NULL
8 H5.3D	H	5	D	Occupied
9 G5.2B	G	5	B	Occupied
10 A4.4B	A	4	B	Occupied
11 A3.1A	A	3	A	Vacant
12 A2.2D	A	2	D	Occupied
13 C1.3A	C	1	A	Occupied
14 E3.2B	E	3	B	Occupied
15 E4.1D	E	4	D	Occupied
16 H3.4C	H	3	C	Occupied
17 A1.1A	A	1	A	Occupied
18 G4.4B	G	4	B	Occupied
19 B2.3A	B	2	A	Occupied
20 A4.3C	A	4	C	Occupied

--table booking

```
INSERT INTO booking VALUES (505, TO_DATE('2024-01-10', 'YYYY-MM-DD'),
TO_TIMESTAMP('12:00:00', 'HH24:MI:SS'), 'CONFIRMED', 2015486, 'G1.2C', 105);
INSERT INTO booking VALUES (532, TO_DATE('2024-01-09', 'YYYY-MM-DD'),
TO_TIMESTAMP('14:30:00', 'HH24:MI:SS'), 'PENDING', 2210923, 'D2.4B', NULL);
INSERT INTO booking VALUES (554, TO_DATE('2024-01-09', 'YYYY-MM-DD'),
TO_TIMESTAMP('14:30:00', 'HH24:MI:SS'), 'CONFIRMED', 2219725, 'H5.3D', 109);
INSERT INTO booking VALUES (515, TO_DATE('2024-01-12', 'YYYY-MM-DD'),
TO_TIMESTAMP('15:30:00', 'HH24:MI:SS'), 'CONFIRMED', 2217611, 'B2.3A', 103);
INSERT INTO booking VALUES (516, TO_DATE('2024-01-11', 'YYYY-MM-DD'),
TO_TIMESTAMP('12:00:00', 'HH24:MI:SS'), 'CONFIRMED', 1926232, 'E3.2B', 102);
INSERT INTO booking VALUES (508, TO_DATE('2024-01-08', 'YYYY-MM-DD'),
TO_TIMESTAMP('10:30:00', 'HH24:MI:SS'), 'PENDING', 2214301, 'A2.2D', NULL);
INSERT INTO booking VALUES (574, TO_DATE('2024-01-13', 'YYYY-MM-DD'),
TO_TIMESTAMP('11:00:00', 'HH24:MI:SS'), 'CONFIRMED', 2228975, 'A1.1A', 104);
INSERT INTO booking VALUES (596, TO_DATE('2024-01-13', 'YYYY-MM-DD'),
TO_TIMESTAMP('14:30:00', 'HH24:MI:SS'), 'CONFIRMED', 1924733, 'A4.3C', 106);
INSERT INTO booking VALUES (555, TO_DATE('2024-01-09', 'YYYY-MM-DD'),
TO_TIMESTAMP('15:00:00', 'HH24:MI:SS'), 'PENDING', 1912346, 'H3.4C', NULL);
INSERT INTO booking VALUES (520, TO_DATE('2024-01-10', 'YYYY-MM-DD'),
TO_TIMESTAMP('09:30:00', 'HH24:MI:SS'), 'PENDING', 2025445, 'E4.1D', NULL);
INSERT INTO booking VALUES (549, TO_DATE('2024-01-10', 'YYYY-MM-DD'),
TO_TIMESTAMP('15:00:00', 'HH24:MI:SS'), 'CONFIRMED', 2312241, 'E2.2A', 109);
```

	BOOKINGID	BOOKING_DATE	BOOKING_TIME	BOOKING_STATUS	STUDENTID	ROOMID	CLEANERID
1	505	10-JAN-24	01-JAN-24	CONFIRMED	2015486	G1.2C	105
2	532	09-JAN-24	01-JAN-24	PENDING	2210923	D2.4B	(null)
3	554	09-JAN-24	01-JAN-24	CONFIRMED	2219725	H5.3D	109
4	515	12-JAN-24	01-JAN-24	CONFIRMED	2217611	B2.3A	103
5	516	11-JAN-24	01-JAN-24	CONFIRMED	1926232	E3.2B	102
6	508	08-JAN-24	01-JAN-24	PENDING	2214301	A2.2D	(null)
7	574	13-JAN-24	01-JAN-24	CONFIRMED	2228975	A1.1A	104
8	596	13-JAN-24	01-JAN-24	CONFIRMED	1924733	A4.3C	106
9	555	09-JAN-24	01-JAN-24	PENDING	1912346	H3.4C	(null)
10	520	10-JAN-24	01-JAN-24	PENDING	2025445	E4.1D	(null)
11	549	10-JAN-24	01-JAN-24	CONFIRMED	2312241	E2.2A	109

6. SQL Query Statements

6.1 Query (select)

i) Display a list of pending bookings for a specific date, showing the student's first and last name, room number, date, and time:

```
SELECT b.studentID, s.student_fname, s.student_lname, b.roomID, b.booking_date,  
b.booking_time  
FROM booking b  
JOIN student s ON b.studentID = s.studentID  
WHERE b.booking_status = 'PENDING'  
AND b.booking_date = TO_DATE('2024-01-09', 'YYYY-MM-DD'); -- Replace with  
the desired date
```

STUDENTID	STUDENT_FNAME	STUDENT_LNAME	ROOMID	BOOKING_D	BOOKING_T
2210923	Hafsa	Halimah	D2.4B	09-JAN-24	01-JAN-24
1912346	Balqis	Raihana	H3.4C	09-JAN-24	01-JAN-24

ii) Select cleaner name and the total number of toilets cleaned by each cleaner and order the results by the total number of toilets cleaned in descending order.

```
SELECT cleaner.cleaner_name, COUNT(*) AS total_toilets_cleaned  
FROM booking  
JOIN cleaner ON booking.cleanerID = cleaner.cleanerID  
GROUP BY cleaner.cleanerID, cleaner.cleaner_name  
ORDER BY total_toilets_cleaned DESC;
```

```

1 row inserted.

CLEANERID CLEANER_NAME                                SHIFT
-----
101 Salmah                                             Morning
102 Rokiah                                             Noon
103 Fatin                                              Evening
104 Saripah                                            Morning
105 Khalida                                            Noon
106 Kamariah                                           Evening
107 Azi                                                Morning
108 Sarina                                             Noon
109 Fida                                               Evening
110 Rahmah                                             Morning
111 Hani                                               Noon

CLEANERID CLEANER_NAME                                SHIFT
-----
478 Alice Smith                                       morning

12 rows selected.

```

6.2 Add (insert)

Insert a new cleaner into the cleaner table:

```
INSERT INTO cleaner (cleanerID, cleaner_name, shift)
```

```
VALUES (478, 'Alice Smith', 'morning');
```

```

1 row inserted.

CLEANERID CLEANER_NAME                                SHIFT
-----
101 Salmah                                             Morning
102 Rokiah                                             Noon
103 Fatin                                              Evening
104 Saripah                                            Morning
105 Khalida                                            Noon
106 Kamariah                                           Evening
107 Azi                                                Morning
108 Sarina                                             Noon
109 Fida                                               Evening
110 Rahmah                                             Morning
111 Hani                                               Noon

CLEANERID CLEANER_NAME                                SHIFT
-----
478 Alice Smith                                       morning

12 rows selected.

```

6.3 Modify (update)

Update the status of a booking to "completed":

UPDATE booking

SET status = 'confirmed'

WHERE bookingID = 532;

1 row updated.						
BOOKINGID	BOOKING_D	BOOKING_T	BOOKING_ST	STUDENTID	ROOMI	CLEANERID
505	10-JAN-24	01-JAN-24	CONFIRMED	2015486	G1.2C	105
532	09-JAN-24	01-JAN-24	confirmed	2210923	D2.4B	
554	09-JAN-24	01-JAN-24	CONFIRMED	2219725	H5.3D	109
515	12-JAN-24	01-JAN-24	CONFIRMED	2217611	B2.3A	103
516	11-JAN-24	01-JAN-24	CONFIRMED	1926232	E3.2B	102
508	08-JAN-24	01-JAN-24	PENDING	2214301	A2.2D	
574	13-JAN-24	01-JAN-24	CONFIRMED	2228975	A1.1A	104
596	13-JAN-24	01-JAN-24	CONFIRMED	1924733	A4.3C	106
555	09-JAN-24	01-JAN-24	PENDING	1912346	H3.4C	
520	10-JAN-24	01-JAN-24	PENDING	2025445	E4.1D	
549	10-JAN-24	01-JAN-24	CONFIRMED	2312241	E2.2A	109
11 rows selected.						

6.4 Remove (delete)

Delete a student record from the student table:

DELETE FROM student

WHERE studentID = 2015486;

(before)

STUDENTID	STUDENT_FNAME	STUDENT_LNAME	CONTACT_NO	ROOMI
2214301	Sabrina	Melati	60173694852	A2.2D
2223692	Dahlia	Alya	60143357445	C1.3A
2123389	Batrisyia	Awatif	60195858632	B2.1A
1920443	Ain	Adlin	60122235456	H5.3D
1926232	Suria	Ilyana	60133546191	E3.2B
2015486	Puteri	Najihah	60102547885	G1.2C
2025445	Aliyah	Fatimah	60135945542	E4.1D
2312241	Adlina	Sofia	60144556744	E2.2A
1912346	Balqis	Raihana	60103235965	H3.4C
2228975	Aisya	Amirah	60136964223	A1.1A
2016583	Alisya	Khadija	60177455854	G4.4B
STUDENTID	STUDENT_FNAME	STUDENT_LNAME	CONTACT_NO	ROOMI
2217611	Maryam	Aminah	60113432556	B2.3A
2210923	Hafsa	Halimah	60111137695	D2.4B
2219725	Nur	Irina	60133566442	H5.3D
1924733	Putri	Sarah	0113255669	A4.3C
15 rows selected.				

(after)

STUDENTID	STUDENT_FNAME	STUDENT_LNAME	CONTACT_NO	ROOMI
2214301	Sabrina	Melati	60173694852	A2.2D
2223692	Dahlia	Alya	60143357445	C1.3A
2123389	Batrisyia	Awatif	60195858632	B2.1A
1920443	Ain	Adlin	60122235456	H5.3D
1926232	Suria	Ilyana	60133546191	E3.2B
2025445	Aliyah	Fatimah	60135945542	E4.1D
2312241	Adlina	Sofia	60144556744	E2.2A
1912346	Balqis	Raihana	60103235965	H3.4C
2228975	Aisya	Amirah	60136964223	A1.1A
2016583	Alisya	Khadija	60177455854	G4.4B
2217611	Maryam	Aminah	60113432556	B2.3A
STUDENTID	STUDENT_FNAME	STUDENT_LNAME	CONTACT_NO	ROOMI
2210923	Hafsa	Halimah	60111137695	D2.4B
2219725	Nur	Irina	60133566442	H5.3D
1924733	Putri	Sarah	0113255669	A4.3C

14 rows selected.

6.5 Remove (delete) with condition

Remove bookings that occurred before a certain date:

DELETE FROM booking

WHERE booking_date < TO_DATE('2024-01-10', 'YYYY-MM-DD');

(before)

BOOKINGID	BOOKING_D	BOOKING_I	BOOKING_ST	STUDENTID	ROOMI	CLEANERID
505	10-JAN-24	01-JAN-24	CONFIRMED	2015486	G1.2C	105
532	09-JAN-24	01-JAN-24	PENDING	2210923	D2.4B	
554	09-JAN-24	01-JAN-24	CONFIRMED	2219725	H5.3D	109
515	12-JAN-24	01-JAN-24	CONFIRMED	2217611	B2.3A	103
516	11-JAN-24	01-JAN-24	CONFIRMED	1926232	E3.2B	102
508	08-JAN-24	01-JAN-24	PENDING	2214301	A2.2D	
574	13-JAN-24	01-JAN-24	CONFIRMED	2228975	A1.1A	104
596	13-JAN-24	01-JAN-24	CONFIRMED	1924733	A4.3C	106
555	09-JAN-24	01-JAN-24	PENDING	1912346	H3.4C	
520	10-JAN-24	01-JAN-24	PENDING	2025445	E4.1D	
549	10-JAN-24	01-JAN-24	CONFIRMED	2312241	E2.2A	109

11 rows selected.

(after)

BOOKINGID	BOOKING_D	BOOKING_I	BOOKING_ST	STUDENTID	ROOMI	CLEANERID
505	10-JAN-24	01-JAN-24	CONFIRMED	2015486	G1.2C	105
515	12-JAN-24	01-JAN-24	CONFIRMED	2217611	B2.3A	103
516	11-JAN-24	01-JAN-24	CONFIRMED	1926232	E3.2B	102
574	13-JAN-24	01-JAN-24	CONFIRMED	2228975	A1.1A	104
596	13-JAN-24	01-JAN-24	CONFIRMED	1924733	A4.3C	106
520	10-JAN-24	01-JAN-24	PENDING	2025445	E4.1D	
549	10-JAN-24	01-JAN-24	CONFIRMED	2312241	E2.2A	109

7 rows selected.

7. Procedure

i) Update Booking Status

```
CREATE OR REPLACE PROCEDURE update_booking_status(
    p_booking_id IN NUMBER,
    p_new_status IN VARCHAR2
)
AS
BEGIN
    UPDATE booking
    SET booking_status = p_new_status
    WHERE bookingid = p_booking_id;

    DBMS_OUTPUT.PUT_LINE('Booking status updated successfully.');
```

EXCEPTION

```
    WHEN NO_DATA_FOUND THEN
        DBMS_OUTPUT.PUT_LINE('Booking ID ' || p_booking_id || ' not
        found.');
```

WHEN OTHERS THEN

```
        DBMS_OUTPUT.PUT_LINE('An error occurred while updating the
        booking status.');
```

END update_booking_status;

/

BEGIN

```
    update_booking_status(532, 'CONFIRMED');
```

END;

/

Before

BOOKINGID	BOOKING_ST
532	PENDING

After

BOOKINGID	BOOKING_ST
532	CONFIRMED

ii) Get room according to status

```
SET SERVEROUTPUT ON
```

```
SET VERIFY OFF
```

```
CREATE OR REPLACE PROCEDURE get_available_rooms(p_room_status IN  
VARCHAR)
```

```
AS
```

```
    CURSOR room_cursor IS
```

```
        SELECT * FROM room
```

```
        WHERE room_status = p_room_status;
```

```
    v_room room%ROWTYPE;
```

```
BEGIN
```

```
    OPEN room_cursor;
```

```
    FETCH room_cursor INTO v_room;
```

```
    DBMS_OUTPUT.PUT_LINE('This is the list of ' || v_room.room_status || ' rooms');
```

```
    WHILE room_cursor%FOUND LOOP
```

```
        -- Process each row as needed
```

```
        DBMS_OUTPUT.PUT_LINE('Room: ' || v_room.roomid);
```

```
        FETCH room_cursor INTO v_room;
```

```
    END LOOP;
```

```
    CLOSE room_cursor;
```

```
END get_available_rooms;
```

```

/
BEGIN
    get_available_rooms('Occupied');
END;

/
BEGIN
    get_available_rooms('Vacant');
END;

/
BEGIN
    get_available_rooms('NULL');
END;

```

Output 1:

```

Procedure GET_AVAILABLE_ROOMS compiled

This is the list of Occupied rooms
Room: B2.1A
Room: G1.2C
Room: E2.2A
Room: H5.3D
Room: G5.2B
Room: A4.4B
Room: A2.2D
Room: C1.3A
Room: E3.2B
Room: E4.1D
Room: H3.4C
Room: A1.1A
Room: G4.4B
Room: B2.3A
Room: A4.3C

PL/SQL procedure successfully completed.

```

Output 2:

```

This is the list of Vacant rooms
Room: H3.3A
Room: D2.4B
Room: A3.1A

PL/SQL procedure successfully completed.

```

Output 3:

```

This is the list of NULL rooms
Room: F1.2A
Room: F2.3C

PL/SQL procedure successfully completed.

```

8. Function

i) Get Student Booking Status

--Function 1: Get student booking status

SET SERVEROUTPUT ON

SET VERIFY OFF

CREATE OR REPLACE FUNCTION get_student_booking_status(

 p_student_id IN NUMBER

) RETURN VARCHAR2

AS

 v_status VARCHAR2(20);

BEGIN

 SELECT b.booking_status

 INTO v_status

 FROM booking b

 JOIN student s ON b.studentid = s.studentid

 WHERE s.studentid = p_student_id;

 RETURN v_status;

```

EXCEPTION

    WHEN NO_DATA_FOUND THEN

        RETURN 'No booking found for student ' || TO_CHAR(p_student_id);

    WHEN OTHERS THEN

        RETURN 'An error occurred while retrieving booking status.';

END get_student_booking_status;

/

ACCEPT v_student_id PROMPT 'Enter student ID: ';

DECLARE

    v_student_id student.studentid%type;

    v_booking_status VARCHAR2(20);

BEGIN

    v_booking_status := get_student_booking_status(&v_student_id);

    DBMS_OUTPUT.PUT_LINE('Student ID: ' || &v_student_id);

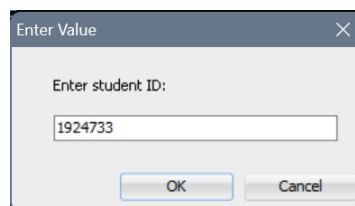
    DBMS_OUTPUT.PUT_LINE('Booking Status: ' || v_booking_status);

END;

/

```

Output



```

Student ID: 1924733
Booking Status: CONFIRMED

PL/SQL procedure successfully completed.

```

ii) Get Cleaner Shift

--Function 2: Get cleaner shift

SET SERVEROUTPUT ON

SET VERIFY OFF

CREATE OR REPLACE FUNCTION get_cleaner_shift(

 p_cleaner_id IN NUMBER

) RETURN VARCHAR2

AS

 v_shift VARCHAR2(20);

BEGIN

 SELECT shift INTO v_shift

 FROM cleaner

 WHERE cleanerid = p_cleaner_id;

 RETURN v_shift;

EXCEPTION

 WHEN NO_DATA_FOUND THEN

 RETURN 'Cleaner with ID ' || TO_CHAR(p_cleaner_id) || ' not found.';

 WHEN OTHERS THEN

 RETURN 'An error occurred while retrieving cleaner shift.';

END get_cleaner_shift;

/

ACCEPT v_id PROMPT 'Enter cleaner ID: ';

DECLARE

 v_cleaner_shift cleaner.shift%type;

 v_id cleaner.cleanerid%type;

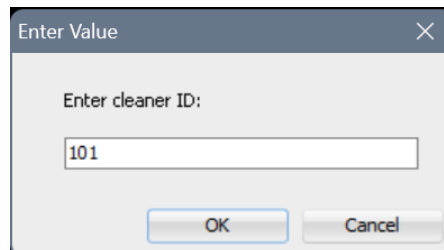
BEGIN

 v_cleaner_shift := get_cleaner_shift(&v_id);

 DBMS_OUTPUT.PUT_LINE('Cleaner ID: ' || &v_id);

```
DBMS_OUTPUT.PUT_LINE('Cleaner Shift: ' || v_cleaner_shift);  
END;  
/
```

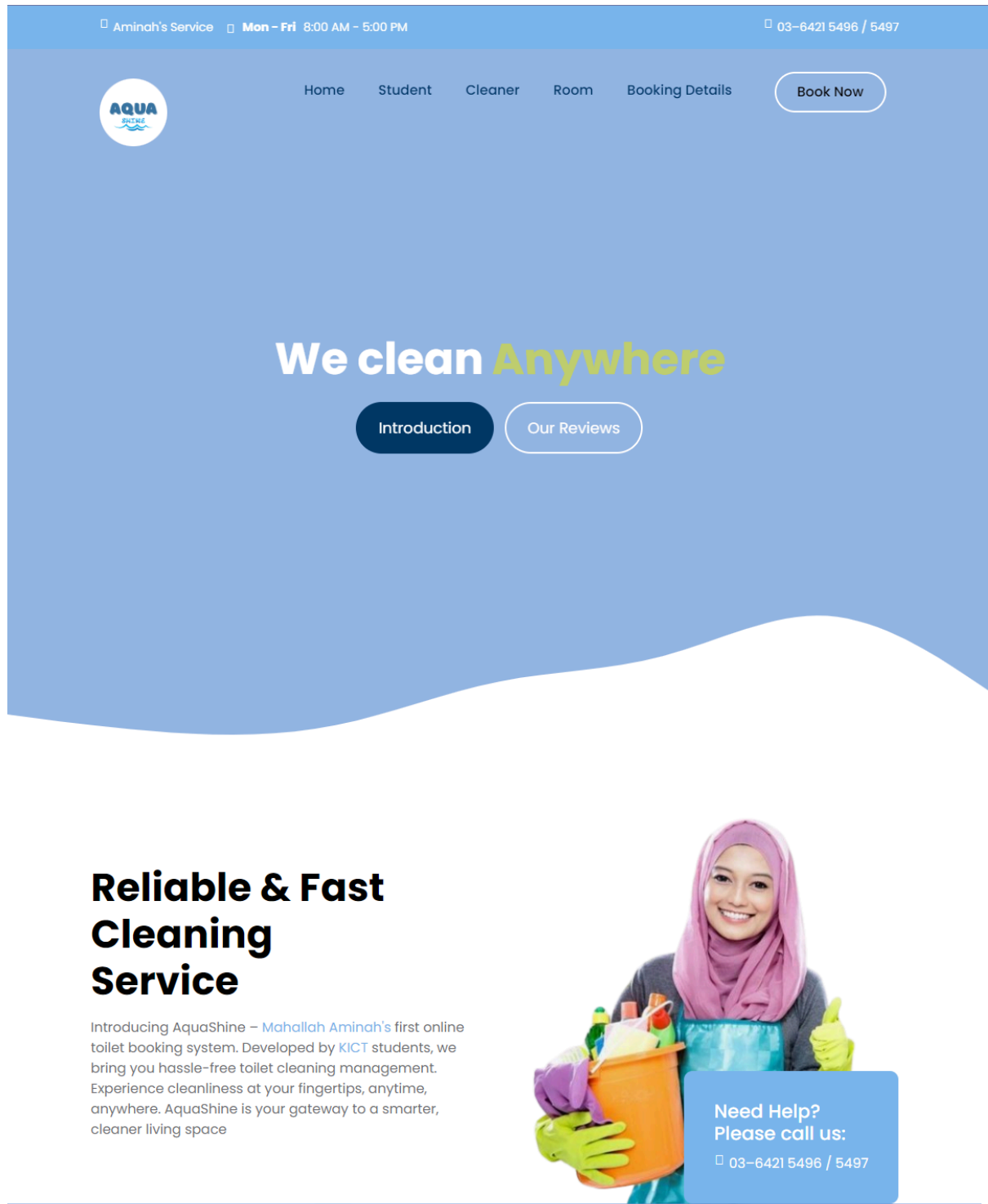
Output



```
Cleaner ID: 101  
Cleaner Shift: Morning  
  
PL/SQL procedure successfully completed.
```

9. Web Application

Homepage



Happy Customers



Sarah

★★★★★

Best sangat guna bila ada website macam ni. Dah tak perlu susah2 book kat whatsapp.



Farah

★★★★★

Easy to use, I can also simply update my status especially when I'm not in my room.



Sofy

★★★★★

Like like like



Nina

★★★★★

Good Service!



Hidayah

★★★★★

Sangat bagus, kerja cepat dan berkualiti. Daebak!



Shai

★★★★★

Overall we were happy with the cleaning. Will definitely recommend your service. Tqsm

Trusted by companies



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Pages

- » [Our Cleaners](#)
- » [Students Data](#)
- » [Room Details](#)
- » [Booking Details](#)

Office

- Mahallah Aminah, International Islamic University Malaysia, Jalan Gombak, 53100 Kuala Lumpur, Malaysia
- 03-6421 5496 / 5497
- mrc_mahallahaminah@iiium.edu.my



Service Hours

Mon - Fri
8:00 AM - 5:00 PM

Student Details

Aminah's Service

Mon - Fri 8:00 AM - 5:00 PM

03-6421 5496 / 5497

AQUA

HomeStudentCleanerRoomBooking DetailsBook Now

STUDENT DETAILS

+ Add New Student

No	Student ID	Student Name		Contact No	Room No	
1	1912346	Adlyn	Adika	60194866884	G2.3D	 
2	1920443	Ain	Adlin	60122235456	H5.3D	 
3	1926232	Suria	Ilyana	60133546191	E3.2B	 
4	2015486	Puteri	Najihah	60102547885	G1.2C	 
5	2016583	Alisya	Khadija	60177455854	G4.4B	 
6	2025445	Aliyah	Fatimah	60135945542	E4.1D	 
7	2123389	Batrisyia	Awatif	60195858632	B2.1A	 
8	2210923	Hafsa	Halimah	6011137695	D2.4B	 
9	2214301	Sabrina	Melati	60173694852	A2.2D	 
10	2217611	Maryam	Aminah	60113432556	B2.3A	 
11	2223692	Dahlia	Alya	60143357445	C1.3A	 
12	2228975	Aisya	Amirah	60136964223	A1.1A	 

Main Menu

Aminah's Service

Mon - Fri 8:00 AM - 5:00 PM

03-6421 5496 / 5497

AQUA

HomeStudentCleanerRoomBooking DetailsBook Now

REGISTER NEW STUDENT

STUDENT ID:

FIRST NAME:

LAST NAME:

CONTACT NO:

ROOM ID:

XXXXXXX

Azni

Murad

601XXXXXXX

F1.4C

ADD NEW STUDENT

STUDENT LIST

Aminah's Service

Mon - Fri 8:00 AM - 5:00 PM

03-6421 5496 / 5497

AQUA

HomeStudentCleanerRoomBooking DetailsBook Now

UPDATE STUDENT RECORD

FIRST NAME:

LAST NAME:

CONTACT NO:

ROOM NO:

Adlyn

Adika

60194866884

G2.3D

UPDATE STUDENT RECORD

STUDENT LIST

Cleaner Details

Aminah's Service

Mon - Fri 8:00 AM - 5:00 PM

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HomeStudentCleanerRoomBooking DetailsBook Now

CLEANER DETAILS

+ Add New Cleaner

No	Cleaner ID	Cleaner Name	Shift	
1	101	Salmah	Morning	 
2	102	Rokiah	Noon	 
3	103	Fatin	Evening	 
4	104	Saripah	Morning	 
5	105	Khalida	Noon	 
6	106	Kamariah	Evening	 
7	107	Azi	Morning	 
8	108	Sarina	Noon	 
9	109	Fida	Evening	 
10	110	Rahmah	Morning	 
11	111	Hani	Noon	 
12	140	Medina	Evening	 
13	153	Didi	Evening	 
14	180	Shila	Evening	 

Main Menu

Aminah's Service

Mon - Fri 8:00 AM - 5:00 PM

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HomeStudentCleanerRoomBooking DetailsBook Now

REGISTER NEW CLEANER

CLEANER ID:

NAME:

SHIFT:

131

Widad

Morning

ADD NEW CLEANER

CLEANER LIST

Aminah's Service

Mon - Fri 8:00 AM - 5:00 PM

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HomeStudentCleanerRoomBooking DetailsBook Now

UPDATE CLEANER RECORD

CLEANER NAME:

SHIFT:

Salmah

Morning

UPDATE CLEANER RECORD

CLEANER LIST

Room Details

Aminah's Service

Mon - Fri 8:00 AM - 5:00 PM

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AQUA

HomeStudentCleanerRoomBooking DetailsBook Now

ROOM DETAILS

+ Add New Room

No	Room ID	Block Name	Level	Compartment	Status	
1	A1.1A	A	1	A	Occupied	
2	A2.2D	A	2	D	Occupied	
3	A3.1A	A	3	A	Vacant	
4	A4.3C	A	4	C	Occupied	
5	A4.4B	A	4	B	Occupied	
6	B2.1A	B	1	A	Occupied	
7	B2.3A	B	2	A	Occupied	
8	C1.3A	C	1	A	Occupied	
9	D2.4B	D	2	B	Vacant	
10	E2.2A	E	2	A	Occupied	
11	E3.2B	E	3	B	Occupied	
12	E4.1D	E	4	D	Occupied	
13	F1.2A	F	1	A	NULL	
14	F2.3C	F	2	C	NULL	
15	G1.2C	G	1	C	Occupied	
16	G4.4B	G	4	B	Occupied	
17	G5.2B	G	5	B	Occupied	
18	H3.3A	H	3	A	Vacant	
19	H3.4C	H	3	C	Occupied	
20	H5.3D	H	5	D	Occupied	

Main Menu

Aminah's Service

Mon - Fri 8:00 AM - 5:00 PM

03-6421 5496 / 5497

AQUA

HomeStudentCleanerRoomBooking DetailsBook Now

ADD NEW ROOM

ROOM ID:

BLOCK:

LEVEL:

COMPARTMENT:

STATUS:

ADD NEW ROOM

ROOM LIST

28

Aminah's ServiceMon - Fri 8:00 AM - 5:00 PM03-6421 5496 / 5497

AQUA

HomeStudentCleanerRoomBooking DetailsBook Now

UPDATE ROOM RECORD

ROOM NO:

A1.1A

BLOCK:

A

LEVEL:

1

COMPARTMENT:

A

STATUS:

Occupied

UPDATE ROOM RECORD

ROOM LIST

Booking Details

Aminah's ServiceMon - Fri 8:00 AM - 5:00 PM03-6421 5496 / 5497

AQUA

HomeStudentCleanerRoomBooking DetailsBook Now

BOOKING DETAILS

+ Add New Booking

No	Booking ID	Date	Time	Status	Student ID	Room ID	Cleaner ID	
1	505	2024-01-10	10:15:04	CONFIRMED	2015486	G1.2C	107	 
2	508	2024-01-08	10:30:00	CONFIRMED	2214301	A2.2D	110	 
3	515	2024-01-12	15:30:00	CONFIRMED	2217611	B2.3A	103	 
4	516	2024-01-11	12:00:00	CONFIRMED	1926232	E3.2B	102	 
5	520	2024-01-10	09:30:00	PENDING	2025445	E4.1D		 
6	532	2024-01-09	14:30:00	PENDING	2210923	D2.4B		 
7	548	2024-01-23	12:03:00	CONFIRMED	2312241	E2.2A	111	 
8	554	2024-01-09	14:30:00	CONFIRMED	2219725	H5.3D	109	 
9	574	2024-01-13	11:00:00	CONFIRMED	2228975	A1.1A	104	 

Main Menu

Aminah's ServiceMon - Fri 8:00 AM - 5:00 PM03-6421 5496 / 5497

AQUA

HomeStudentCleanerRoomBooking DetailsBook Now

INSERT NEW BOOKING

BOOKING ID:

5XX

DATE:

mm/dd/yyyy

TIME:

--:--:--

STUDENT ID:

ROOM NO:

CLEANER ID:

ADD NEW BOOKING

BOOKING LIST

Aminah's Service
Mon - Fri 8:00 AM - 5:00 PM
03-6421 5496 / 5497



[Home](#)
[Student](#)
[Cleaner](#)
[Room](#)
[Booking Details](#)
[Book Now](#)

UPDATE BOOKING RECORD

CHANGE DATE:

01/10/2024

CHANGE TIME:

10:15:04 AM

STATUS:

CONFIRMED

STUDENT ID:

2015486

ROOM NO:

G1.2C

CLEANER ID:

107

UPDATE BOOKING RECORD

BOOKING LIST

10. Member Contributions

Task	Name	Matric No
1. Introduction	Adlyn Sofea	2214042
2. Entity Relationship Diagram (ERD)	Wardah Fatimah	2210902
3. Data Dictionary	Adlyn Sofea	2214042
4. Data Definition Language (DDL) - Create Table	Bidayatun Najihah	2216798
5. Data Definition Language (DDL) - Data Population	Wardah Fatimah	2210902
6. SQL Query Statements	Bidayatun Najihah	2216798
7. Procedure	Nur Alifah Ilyana	2217466
8. Function		
9. Web Application	Nur Alifah Ilyana	2217466
	Adlyn Sofea	2214042